

motherboard socket. You will see markings like A, B, C, D, A1, A2, B1, B2 etc right beside each RAM slot. These help decipher the different memory configurations on your motherboard. In order for your memory to work in Dual channel or Quad channel (whichever is supported by your motherboard) you need to use certain slots and leave the others empty. These configurations are mentioned in your motherboard manual along with all the different memory modules that your computer was tested with.

3. North Bridge / South Bridge / Platform Controller Hub

The earlier motherboard architecture had a structure in which the functions and the control of the various motherboard components were divided amongst the SB (South Bridge) and the NB (North Bridge). The North Bridge primarily controlled the communication between the RAM + GPU with the CPU. All communication to the various peripherals was via the SB which was relayed to the CPU by the NB. However, ever since Sandy Bridge came out 2 years ago and AMD's APU came out around the same time, this approach has changed. This was primarily due to the fact that the functions performed by the North Bridge were incorporated onto the processor. This greatly reduced latency and allowed for faster communication.

The South Bridge was connected to the North Bridge and relayed to the processor whatever the various peripherals had to say. Since the migration of the North Bridge onto the processor, the South Bridge has been replaced by the PCH (Platform Controller Hub) which now connects directly to the processor. All USB ports, some SATA ports and other peripherals like FireWire, COM port, legacy ports etc are now controlled by the PCH. Now the PCH depends on the model of the CPU, there are multiple PCH chips with varying capabilities. Certain PCH allow overclocking, extra USB 2.0 / 3.0 ports, extra PCIe slots or maybe even some features like SSD caching. So it is advisable to figure out what features are important to you before buying a motherboard, everything else depends on it.

4. Power Management

Power Phases are components on your motherboard that take power from the SMPS and then regulate it before supplying your CPU and RAM. The phase circuitry is composed of VRMs and MOSFETs. A regulated power supply is very important for your CPU since that determines how much punishment your processor can take. Without a good power phase design